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$\Rightarrow \quad 4x^2 = (1-x)^2$

$\Rightarrow \quad 4x^2 = 1 + x^2 - 2x$

$3x^2 + 2x - 1 = 0$

$$x = \frac{-2 \pm \sqrt{4 - 4 \times 3 \times -1}}{2(3)}$$

$$x = \frac{-2 \pm \sqrt{4+12}}{6}$$

$$= \frac{-2 \pm \sqrt{16}}{6}$$

$$= \frac{-2+4}{6} ; \frac{-2-4}{6}$$

$$= \frac{2}{6} ; \frac{-6}{6}$$

$x = 0.33 ; -1$ (not possible)

$\therefore [A_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[B_2]_{eq} = 1 - x = 1 - 0.33 = 0.67$

$[AB]_{eq} = 2x = 2 \times 0.33 = 0.66$

50. $K_{P_1} = 8.19 \times 10^2 \quad T_1 = 298 \text{ K}$

$K_{P_2} = 4.6 \times 10^{-1} \quad T_2 = 498 \text{ K}$

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^o}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{4.6 \times 10^{-1}}{8.19 \times 10^2}\right) = \frac{\Delta H^o}{2.303 \times 8.314}\left(\frac{498 - 298}{498 \times 298}\right)$$

$$\frac{-3.2505 \times 2.303 \times 8.314 \times 498 \times 298}{200} = \Delta H^o$$

$\Delta H^o = -46181 \text{ J mol}^{-1}$

$\Delta H^o = -46.18 \text{ KJ mol}^{-1}$

51. $P_{CO_2} = 1.017 \times 10^{-3} \text{ atm} \quad T = 500^o \text{ C}$

$K_P = P_{CO_2}$

$\therefore K_{P_1} = 1.017 \times 10^{-3} \quad T = 500 + 273 = 773 \text{ K}$

$K_{P_2} = ? \quad T = 600 + 273 = 873 \text{ K}$

$\Delta H^o = 181 \text{ KJ mol}^{-1}$

$$\log\left(\frac{K_{P_2}}{K_{P_1}}\right) = \frac{\Delta H^o}{2.303\,R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3}{2.303 \times 8.314}\left(\frac{873 - 773}{873 \times 773}\right)$$

$$\log\left(\frac{K_{P_2}}{1.017 \times 10^{-3}}\right) = \frac{181 \times 10^3 \times 100}{2.303 \times 8.314 \times 873 \times 773}$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = \text{anti log of } (1.40)$$

$$\frac{K_{P_2}}{1.017 \times 10^{-3}} = 25.12$$

$\Rightarrow K_{P_2} = 25.12 \times 1.017 \times 10^{-3}$

$K_{P_2} = 25.54 \times 10^{-3}$

Unit - 9

Evaluate yourself :

1) mass of KOH = 5.6 g

no. of moles = $\frac{5.6}{56}$ = 0.1 mol

i) Volume of the solution = 500 ml = 0.5 L

ii) Volume of the solution = 1 L

$$\text{molarity} = \frac{\text{number of moles of solute}}{\text{volume of solution (in L)}}$$

$$= \frac{0.1}{0.5} = 0.2 \text{ M}$$

ii) Volume of the solution = 1 L

$$\text{molarity} = \frac{0.1}{1} = 0.1\text{M}$$

2. mass of glucose = 2.82 g

no. of moles of glucose = $\frac{2.82}{180} = 0.016$

mass of water = 30 g